Tamil Nadu - Kerala), Nilgiri (Tamil Nadu - Kerala), Gulf of Mannar (Tamil Nadu) are the BRs notified in Tamil Nadu.

Sacred Groves

A sacred grove or sacred woods are any grove of trees that are of special religious importance to a particular culture. Sacred groves feature in various cultures throughout the world. The examples of sacred groves are Alagar Hills in Tamil Nadu, Khasi and Jaintia Hills in Meghalaya and Aravalli Hills of Rajasthan.

11.7.2 *Ex-Situ* Conservation

It is conservation of selected rare plants/ animals in places outside their natural homes. It includes offsite collections and gene banks.

Offsite Collections

They are live collections of wild and domesticated species in Botanical gardens, Zoological parks, Wildlife safari parks, Arborata (gardens with trees and shrubs). The organisms are well maintained for captive breeding programmes. As a result, many animals which have become extinct in the world continue to be maintained in Zoological Parks. As the number increases in captive breeding, the individuals are selectively released in the wild. In this way the Indian crocodile and gangetic dolphin have been saved from extinction.

Gene Banks

Gene banks are a type of biorepository which preserve genetic materials. Seeds of different genetic strains of commercially important plants can be stored in long periods in seed banks, gametes of threatened species can be preserved in viable and fertile condition for long periods using cryopreservation techniques.

However, it is not economically feasible to conserve all biological wealth and all the ecosystems. The number of species required to be saved from extinction far exceeds the conservation efforts.

Table 11.4 Difference between *Insitu* and *Exsitu* Conservation

***Insitu* Conservation**	***Exsitu* Conservation**
It is the on-site conservation or the conservation of genetic resources in natural populations of plant or animal species.	This is a conservation strategy which involves placing of threatened animals and plants in special care locations for their protection.
It is the process of protecting an endangered plant or animal species in its natural habitat, either by protecting or restoring the habitat itself, or by defending the species from predators.	It helps in recovering populations or preventing their extinction under simulated conditions that closely resemble their natural habitats.
National Parks, Biosphere Reserve, Wild Life Sanctuaries form *insitu* conservation strategies.	Zoological parks and Botanical gardens are common *exsitu* conservation programs.

Summary

Biodiversity is the variety of all life on Earth, encompassing genetic, species and ecosystem diversity. Today's biodiversity is the fruit of billions of years of evolution, shaped by natural processes and, increasingly, by the influence of humans. To date, about two million species have been identified on Earth.

Biodiversity supplies a large number of goods and services that sustain human life, including the provision of food, fuel and building materials; purification of air and water; stabilization and moderation of the earth's climate; moderation of floods, droughts, temperature extremes and wind forces; generation and renewal of soil health; maintenance of genetic resources as inputs to crop varieties and livestock breeds, medicines, and other products; and cultural, recreational and aesthetic benefits.

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